Sahayak

ABV IIITM STUDENT BRANCH IEEE MP SECTION

HACKSAGON

A NATIONAL LEVEL SOFTWARE & HARDWARE HACKATHON

Software Track : Open Innovation

Topic: Sahayak-A Helping Hand

Team: Midnight-Merge

"Clear access, real support — empowering persons with disabilities to navigate confidently"

"Before we begin, think about this:"

~ "A step for you. A barrier for millions."

For millions in India, going to a hospital, bank, or exam center starts with fear — 'Is the entrance even accessible?'"

"2.68 crore Indians face this uncertainty every single day."

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Problem Statement

In India, persons with disabilities (PwDs) face daily uncertainty while visiting public places such as hospitals, banks, offices, and exam centers.

There is no reliable, real-time system to check:

- Whether an entrance has ramps or steps
- If the width is wheelchair-accessible
- Presence of obstacles or heavy crowd
- Availability of on-ground assistance

Because of this:

- PwDs experience fear, confusion, and dependence
- Many visits result in wasted travel, stress, or cancellation
- Accessibility exists in policy, but not in practical awareness

“ 2.68 crore Indians face this challenge every day.”

Hence, there is a strong need for a solution that provides clear, real-time accessibility information *before* visiting a place.

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PROPOSED SOLUTION

Sahayak – A Helping Hand is a digital accessibility assistant designed for persons with disabilities (PwDs). It provides **verified, real-time accessibility information** of public places **before the user visits and** connect with **verified volunteer** when they need any kind of help.

The platform:

- Analyzes entrances using **AI-based image detection**
- Shows **clear accessibility indicators** (ramps, steps, width, obstacles)
- Uses **community confirmation** for real-time accuracy
- Connects users with **nearby volunteers** when physical assistance is needed
- This removes uncertainty, fear, and dependency, enabling **independent decision-making**.

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Tech Stack

- **React / React Native** : Cross-platform support (Android + Web)
- **Firebase / Firestore** : Real-time updates for community responses
- **Node.js + Express** : Scalable backend for API handling
- **YOLO / MobileNet** : Fast and accurate detection of ramps, steps, entrances
- **OCR (Tesseract)** : Reads notices/signboards for accessibility info
- **Firebase Auth / OTP** : Simple & inclusive login for PwDs
- **Grok API / Gemini API:** Assists in intelligent analysis, suggestions, and future **expansion** like smart guidance and recommendations

- This stack ensures **speed, scalability, and accessibility-first design**.

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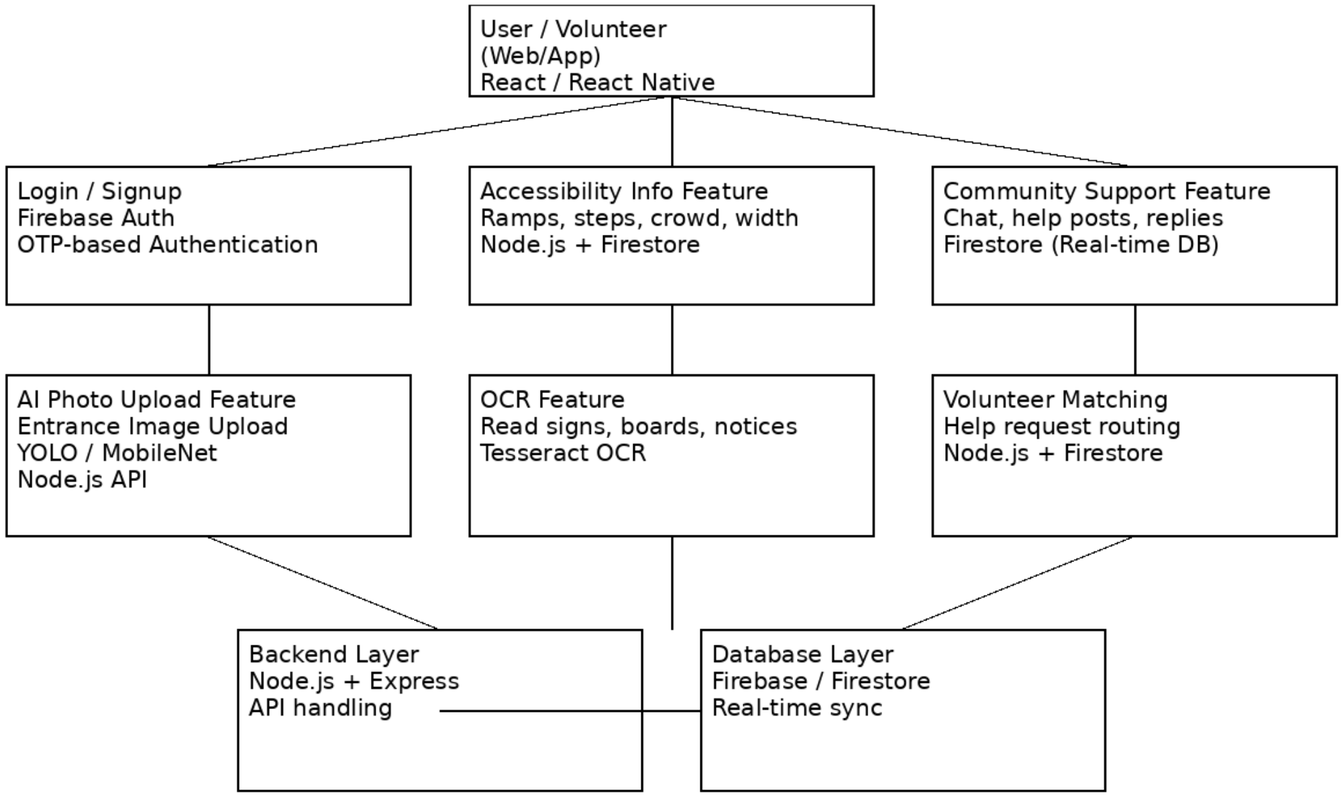
FLOWCHART



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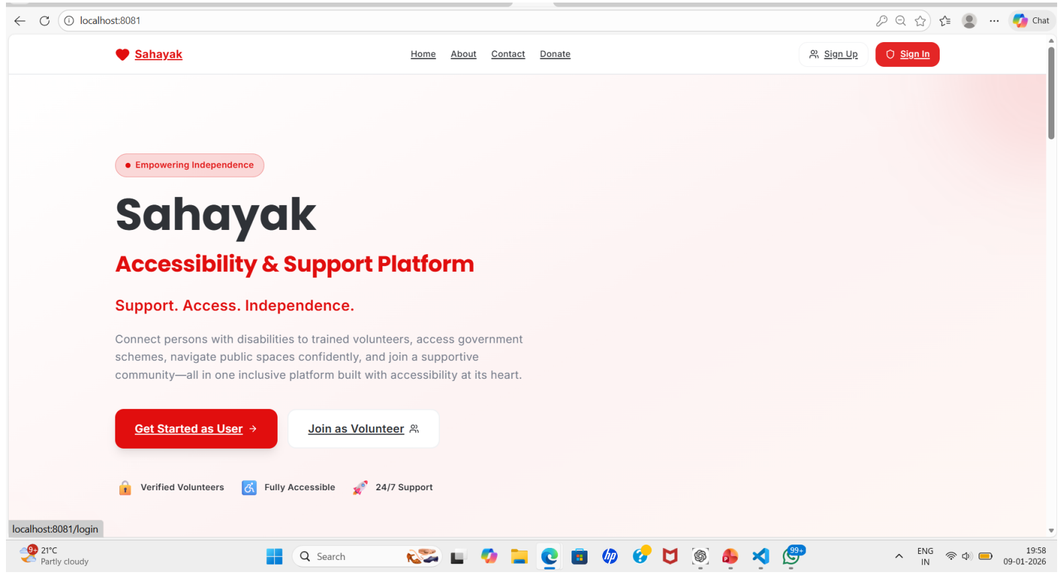
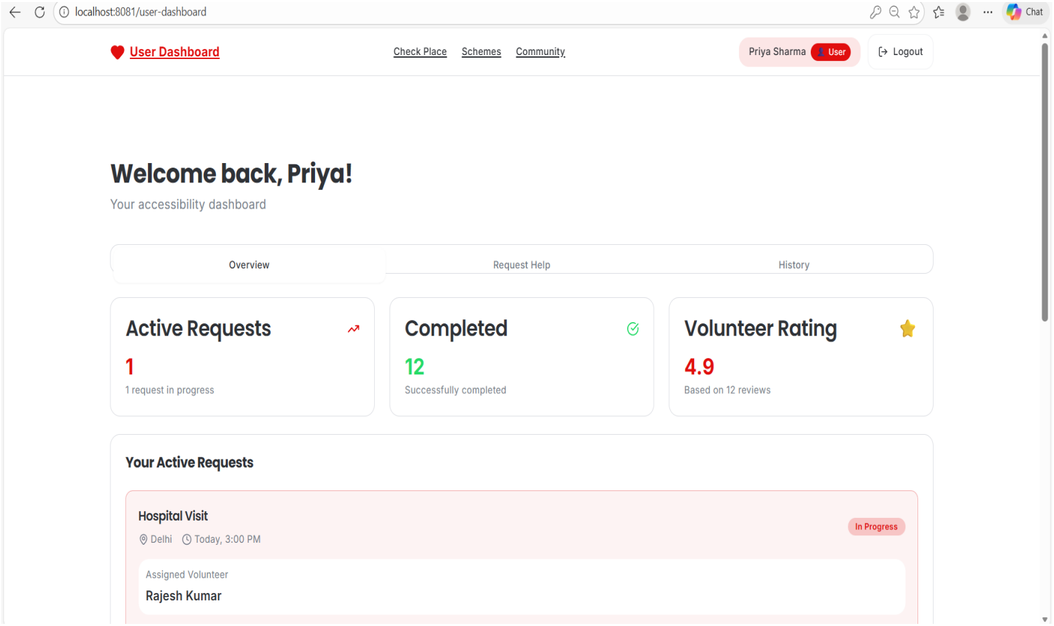
WORKING PROTOTYPE

Prototype Video : https://drive.google.com/file/d/1w-YD-qMK-S5WagfJNVCcS0vxUsztph-k/view?usp=sharing





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FEATURES AND NOVELTY

Core Features

- AI-based ramp & step detection
- Real-time accessibility status
- Community verification system
- Volunteer support for on-ground help
- Simple UI designed for PwDs

Novelty / Uniqueness

- Focuses on pre-visit clarity, not post-visit complaints
- Combines AI + human validation
- Works even where government data is missing
- Built specifically for Indian infrastructure realities

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IMPACT POTENTIAL

Social Impact

- **Inclusive participation:** Enables 2.68 crore PwDs in India to access education, jobs, exams, and public services confidently.
- **Real-time accessibility awareness:** Crowdsourced data helps communities and authorities improve ramps, lifts, and entry points.
- **Reduced isolation:** Connects PwDs with verified volunteers → builds trust and community support.
- **Better scheme utilization:** Simplified guidance helps PwDs actually *use* government benefits, not just know about them.

Economic Impact

- **Higher workforce participation:** Improved mobility = more employability → boosts local productivity.
- **Lower dependency costs:** Reduces caregiver time, travel waste, and last-minute cancellations.
- **Institutional benefits:** Offices, hospitals, colleges get accessibility feedback → fewer complaints, better compliance.
- **Youth empowerment:** Volunteers gain certificates + real-world experience → stronger resumes + social sector careers.

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DRAWBACK AND SHOWSTOPPERS

Drawbacks

- Initial lack of user-generated data
- Dependence on community participation
- AI accuracy may vary with poor image quality

Showstoppers & Mitigation

- **Low adoption initially** → Tie-ups with NGOs & colleges
- **False information risk** → Community verification + reporting
- **Volunteer safety** → Verified profiles & limited assistance scope

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Name of team members & their contact details:

Team Member Name	Contact number
Anshumala Pandit(Team Leader)	8263886589
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THANK YOU

“Development feels complete only when its reaches to everyone”